MILITARY SPECIFICATION

# FILTER UNIT, INTRAVENOUS SOLUTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of intravenous solution filter unit that provides final filtering of particulates which may be present in intravenous solutions.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-585 | Boxes, Wood, Wirebound. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Box, Fiberboard. |

Military

| | |
|---|---|
| MIL-L-10547 | Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible. |
| MIL-S-36586 | Sterilization Strip Test, Bacterial Spore. |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |



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(Copies of specifications, and standards, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (USP)

(Application for copies shall be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042.)

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE:
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and General Regulations for its Enforcement.

(Application for copies should be addressed to Food and Drug Administration, U.S. Department of Health, Education, and Welfare, Washington, D.C. 20203.)

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

| | |
|---|---|
| ANSI Z70.1 | American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications. |

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Rubber and plastic components. All rubber and plastic components shall be of the best quality and suitable in all respects for the intended purpose. All rubber and plastic components shall be sterilizable, free from pyrogens, compatible, and safe (nontoxic) when tested as specified in section 4. Those rubber and plastic components coming in contact with solutions being administered, and end protectors, shall contain not more than 5 parts per million (p.p.m.) total of lead, barium, and cadmium when tested as specified in 4.4.1, or 4.4.1.1. The plastics shall meet the requirements of the Pharmacopeia of the United States (USP) for Class II plastics and for transfusion and infusion assemblies when tested as specified in 4.4.2.



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3.1.1.1 Medication injection site. The medication injection site shall be fabricated from latex.

3.1.1.2 Plastic tubing. Plastic tubing shall be clear, transparent, flexible polyvinyl chloride. The tubing shall be flexible and pliable, but sufficiently rigid to prevent collapse during normal use. The tubing shall be clean and free of particles and shall be free from all liquid lubricating substances which oxidize, exude, or sweat-out to the surface of the tubing.

3.1.2 Other material. All other material hereinafter specified shall be of the best quality and suitable in all respects for the purpose intended.

3.2 Style, design and construction. Shall be a disposable, transparent, flexible, plastic, self contained, solution additive inline final filter unit which shall retain particles larger than 0.5 micron. The overall length of filter set shall be a minimum of 9 inches terminating at one end and by attachment to a tapered male adapter and at the other end by a female adapter. Shall be supplied with a shut off clamp and medication injection site.

3.2.1 Filter. The filter shall consist of the appropriate filter membrane sealed between transparent rigid plastic top and bottom sections. The top or inlet section shall contain a female Luer taper in accordance with Figure 3 of ANSI Standard Z70.1. The bottom or outlet section shall contain a fitment to receive plastic tubing in a securely attached leakproof manner. The fit of the Luer cavity of the female adapter shall be tested by means of the plug gage (without feeler gage) as per Figure 5 of ANSI Standard Z70.1, with a notch (denoting tolerance) extending for the entire 0.057) inch thickness of the feeler gage which includes the "GO" - "NO GO" thickness. As per Figure 5, with the feeler gage removed, the fit of the female Luer cavity shall not bottom out on the plug gage, and must be within a tolerance including the "GO" - "NO GO" dimensions of the feeler gage.

3.2.2 Male adapter. The male adapter shall be fabricated from a rigid plastic and shall be approximately 1-1/2 inch in overall length. The taper shall meet generally the requirements of the American National Standards Institute (ANSI) Standard Z70.1 and shall form a tight leakproof fit with needle hub conforming with the requirements of ANSI Standard Z70.1.

3.2.3 Medication injection site. The rubber part shall be capable of resealing subsequent to puncture with a nonsiliconized 18 gage needle. When tested for resealing as specified in 4.4.2, there shall be no falling drop within 10 seconds.

3.2.4 Control clamp. Control clamp shall be a metal slide clamp fabricated from aluminum. When the clamp is operated from its fully open position to complete shutoff, the tubing shall not be cut, pierced or damaged so as to cause leakage or restriction of flow of solution when tested in accordance with 4.4.3.



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3.3 Sterility. The filter unit shall be sterile when tested with a biological indicator as specified in 4.4.

3.3.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa., 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.3.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.4 Safety (toxicity). All equipment shall be safe. When tested as specified in 4.4.2, the filter unit shall comply with requirements for safety (toxicity) of the USP.



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3.5 Freedom from pyrogens. The fluid path and end connectors shall be pyrogen-free when tested as specified in 4.4.3.

3.6 Certificates of freedom from pyrogen, safety, and biological tests. A certificate listing all lot or control numbers shall state that representative samples from each lot or control have been tested and found to be free from pyrogens, nontoxic (safe), and have passed the biological tests. This certificate shall include the specific tests performed and the test results. For destination inspection, certificates shall be submitted by the contractor directly to Defense Personnel Support Center, Directorate of Medical Materiel, Technical Operations Division, 2800 South 20th Street, Philadelphia, Pa. 19101, Attention: Quality Assurance Branch, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection, certificates shall be furnished to the procuring activity via the government inspector.

3.7 Manufacturer's record. The manufacturer shall place a code or lot number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all tests made on nontoxic, sterile, pyrogen free and biological tests performed on the kits. These records shall be made available for inspection by any duly authorized Government representative. No shipment shall be made until all tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.8 Workmanship. Workmanship shall be first class throughout. Filter units shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.



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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table I.

Table I. Sampling for examination

| Defect category | Inspection level | AQL (percent defective) |
|---|---|---|
| Major A | S-2 | 1.0 |
| Major B | S-2 | 1.0 |
| Minor | S-2 | 1.5 |

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table III.

Table II. Sampling Criteria for Tests

| Component | Characteristic | Requirements | Test procedure | Inspection level | AQL (percent defective) |
|---|---|---|---|---|---|
| Filter Units | Sterility | 3.3 | 4.4.1 | ** | * |
| | Safety | 3.4 | 4.4.2 | ** | * |
| | Freedom from pyrogens | 3.5 | 4.4.3 | ** | * |

* Acceptance number shall be zero for all sample sizes.
**Sampling for tests shall be on component lots.

4.2.3 Sample unit usage. Where possible, the same sample used in the construction of the filter unit shall be used for the determination of two or more test characteristics.



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4.3 Examination. The filter unit shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in Table III. Examination shall not be restricted to the classified possible defects listed in Table III. Any filter unit in the sample containing one or more defects shall be rejected and, if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

Table III. Classification of defects

| Categories | Defects |
|---|---|
| Critical | None defined. |
| Major A | |
| 101 | Filter unit incomplete, one or more components missing. |
| Major B | |
| 151 | Metal or plastic parts not smoothly finished; not free of rough edges, slivers, nicks, pits, blisters, scratches, cracks, bubbles, roughness, or corrosion, as applicable. |
| 152 | Style and design of components not as specified. |
| Minor | |
| 201 | Packaging, packing and marking not as specified. |
| 202 | Poor workmanship (not resulting in a major defect). |

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Sterility test. The filter unit shall be tested for sterility in accordance with the method specified in the USP for all components of the filter unit using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.4.1.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in Table II, Sampling Criteria for Tests.



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4.4.1.1 Inoculated product or simulated product for each type of sterilization, the organism used shall be as listed below:

| Type of Sterilization | Organism | Incubation Temperature |
|---|---|---|
| Steam | B. stearothermophilus | 55° - 65°C |
| Dry Heat | B. stearothermophilus | 55° - 65°C |
| | B. subtilis var. niger (or globigii) | 30° - 35°C |
| Gas | B. subtilis var. niger (or globigii) | 30° - 35°C |
| Radiation | B. pumilus | 30° - 35°C |

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.1.2 Inoculated carrier. Inoculated carrier (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.1 Sterility test and 4.4.1.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.1.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.1.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.1.4 Product testing. Testing for sterility of the product after receipt by the Government shall be without a biological indicator and in accordance with the USP.

4.4.2 Safety. The filter unit shall be tested for safety in accordance with the USP.

4.4.3 Freedom from pyrogens. Samples of the fluid path and end connectors shall be tested for freedom from pyrogens, in accordance with the USP.



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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package containing forty eight filter units, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Individual package. Each filter unit shall be packaged in a sealed tamperproof container capable of maintaining sterility of contents unless damaged or opened. The container shall be a combination polyethylene and paperback, peel open type bag. The container shall be designed with tabs to facilitate opening. As an alternate, the container may be a paperboard tray with a clear plastic wrap designed with a tear strip to facilitate opening. The filter unit shall be coiled in such a manner so as not to kink or be permanently deformed.

5.1.2.2 Unit package. Forty eight individually packaged filter units shall be packaged in a snug fitting box constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be secured to prevent spilling of contents under normal handling.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Levels A, B or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Four unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification. Fiberboard boxes shall conform to the special requirements as specified in PPP-B-636.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with paragraph 30.4 of PPP-B-636.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.



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5.2.2 Level B.

5.2.2.1 Exterior container. Four unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Fiberboard boxes shall conform to special requirements of Table I of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.4 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.3 Marking.

5.3.1 Individual package. In addition to any commercial markings, each individual package shall be marked in accordance with the requirements of the Federal Food Drug and Cosmetic Act. In addition, marking shall include the lot or control number, name of contents, name and address of the contractor and the legends or similar wordings:

"STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED"
"CAUTION: DO NOT ALLOW DIAGNOSTIC DYES TO REMAIN IN TUBING LONGER THAN TWO HOURS"
"COMPATIBLE WITH SODIUM METHIODAL, MEGLUMINE DIATRIZOATE INJECTION USP, SODIUM ACETRIZOATE INJECTION USP AND SODIUM DIATRIZOATE INJECTION USP."

5.3.2 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number and name and address of contractor shall be shown. The method of sterilization (G-Gas, R-Radiation, C-Cold, S-Steam, D-Dry Heat Sterilization) shall be shown following the level of packaging and the date of packaging. Example: "A-10/74-G". Final sterilization shall be on or after indicated packaging date.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown. The method of sterilization as specified in 5.3.2 shall also be shown.



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6. NOTES

6.1 Intended use. The filter unit is intended for use in providing final filtering of particulates which may be present in intravenous solution.

6.2 Ordering data. The procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

| National Stock No. | Item Identification |
|---|---|
| 6515-00-135-2160 | FILTER UNIT, INTRAVENOUS SOLUTION, Fine Pore, 48s |

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1883

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-F-37101

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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